

Paper 7 Candidate A Design Log v0.1

Operator-Layer Cost Compression, Inventory Formation, and Non-Selection Evidence

A Boundary Model for Backend Retained Demand

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May 2026

Retained-Demand Audit Series — Phase II Materials

Abstract

This design log preserves a possible future research path for Paper 7 Candidate A in the Retained-Demand Audit Series. It is not a full paper draft and does not launch Paper 7. The document develops a boundary model for backend retained demand after user abstraction, asking when infrastructure operators must move from just-in-time sourcing to pre-positioned inventory, collateral, liquidity buffers, or fallback liquidity lines.

The central question is simple: when users no longer see or hold the asset, who must hold it in the background? The note distinguishes capability from demand, customer utility from asset necessity, usage from inventory, and alternative possibility from alternative superiority.

The design log introduces three linked audit concepts: the JIT-versus-inventory boundary, the Inter-Zone Liquidity Boundary Model, and the Alternative Infrastructure Maintenance Cost Lens. It also preserves non-selection evidence as a methodological safeguard: if an asset is technically and institutionally available but is not held by operators despite favorable boundary conditions, that absence may become evidence against the retained-demand hypothesis.

XRP is used only as a stress-test case. The note does not argue that XRP must be selected, nor does it treat Ripple-related infrastructure growth as sufficient evidence of XRP retained demand. The framework is intended to define the evidence conditions under which backend retained demand may appear, and the conditions under which persistent non-selection should lower the hypothesis.

What This Is Not

This is not Paper 7, not a working paper, not an investment thesis, and not a claim that XRP or any specific asset currently satisfies the retained-demand conditions described here.

This is a design log for preserving audit questions, boundary conditions, and future research paths. It does not identify a winner.

Why This Note Matters

The central question of this note is simple:

When users no longer see or hold the asset, who must hold it in the background?

After user abstraction, visible usage may disappear from the interface while operational burden moves to routers, market makers, liquidity providers, custodians, exchanges, prime brokers, payment operators, or treasury systems.

This note preserves a boundary model for that shift. It asks when just-in-time sourcing is enough, when pre-positioned inventory becomes rational, and when persistent non-selection should count against a retained-demand hypothesis.

Status Note

This document is not a full paper draft and does not launch Paper 7.

It records the design logic, terminology, boundary conditions, and audit posture developed after Paper 6 of the Retained-Demand Audit Series, After User Abstraction: Backend Retained Demand and Just-in-Time Liquidity in Institutionally Connected Digital Assets.

For citation or archival purposes, this document should be referenced as a Phase II design log within the Retained-Demand Audit Series, not as Paper 7.

Scope and Claim Boundaries

This note is a conceptual design log, not an empirical finding, investment thesis, valuation model, or prediction.

It makes no claim that any specific asset currently satisfies the retained-demand conditions described here. It also makes no claim that XRP, RLUSD, Ripple-related systems, stablecoin rails, tokenized deposits, bank APIs, Canton-like networks, or other institutional infrastructures will necessarily dominate future settlement architecture.

The claims in this note are limited to the design of an audit framework:

1. how to distinguish usage from retained demand;
2. how to distinguish user-facing utility from backend asset necessity;
3. how to compare just-in-time sourcing with pre-positioned inventory;
4. how to compare asset-inclusive configurations with asset-free composite infrastructure;
5. how to treat persistent non-selection as potentially meaningful evidence.

1. Purpose

Paper 6 examined how user abstraction changes the location of retained demand. In a user-abstracted environment, end users may no longer directly hold or manage bridge assets, gas tokens, settlement assets, or jurisdiction-specific payout instruments. Instead, routers, paymasters, liquidity providers, market makers, custodians, exchanges, prime brokers, payment operators, and treasury systems may perform those functions in the background.

This design log extends that inquiry by asking a more specific question:

After an asset disappears from the user interface, under what conditions must infrastructure operators hold that asset as inventory, collateral, margin, a liquidity buffer, or a fallback liquidity line?

The purpose of this document is not to launch Paper 7, but to preserve the design logic for a possible future paper on the boundary between just-in-time sourcing and pre-positioned inventory after user abstraction.

2. Core Thesis

The core thesis of this design log is deliberately conditional. It does not assert that backend retained demand exists for any specific asset. It defines the conditions under which such demand may be inferred.

Backend retained demand after user abstraction does not arise merely because an asset can be used, routed, displayed, or supported. It arises only when infrastructure operators must hold the asset because holding it reduces total operator cost, execution failure risk, market impact, delay, liquidity fragmentation, operational complexity, or reachability failure relative to just-in-time sourcing or asset-free composite infrastructure.

Use without inventory is weak retained demand. Inventory appears only when holding beats sourcing.

This distinction prevents usage, infrastructure adjacency, customer utility, and technological capability from being prematurely converted into evidence of asset-level retained demand.

3. From User Abstraction to Backend Retained Demand

User abstraction changes the visible location of asset interaction. End users may experience a simple payment, treasury, custody, settlement, or application interface without seeing the underlying asset path. They may not know which token, stablecoin, chain, deposit token, bank rail, or liquidity venue is used in the background.

However, abstraction does not eliminate operational burden. It relocates it. Someone may still need to source liquidity, hold inventory, maintain collateral, manage margin, provide payout access, absorb execution failure, handle liquidity fragmentation, guarantee settlement timing, maintain fallback paths, and satisfy compliance, accounting, custody, and risk controls.

The key question is therefore not whether users hold the asset. The key question is whether infrastructure operators must hold the asset on behalf of the user-facing system.

In a user-abstracted system, retained demand may appear not in the user wallet, but in the balance sheets, liquidity systems, or risk-management structures of backend operators.

4. Core Distinctions

This design log relies on four distinctions that should remain explicit in any future paper derived from it.

Distinction	Meaning	Audit implication
Capability does not imply demand	An asset may be supported, stored, routed, bridged, displayed, tokenized, or integrated.	Capability is technical or operational readiness; it is not retained demand.
Customer utility does not imply asset necessity	A system may deliver strong customer utility without requiring a specific asset to be retained.	Asset necessity must be tested by removal sensitivity.
Usage does not imply inventory	An asset can be used transiently without being retained.	Retained demand requires inventory, collateral, margin, buffers, or removal sensitivity.
Alternative possibility does not imply alternative superiority	A system may bypass a candidate asset but still recreate other maintenance costs.	The correct comparison is total operator cost, risk, and complexity.

The correct comparison is not whether an asset can be bypassed. The correct comparison is whether the asset-free composite infrastructure is cheaper, safer, deeper, simpler, and more reliable than the asset-inclusive configuration.

5. JIT Sourcing Versus Pre-Positioned Inventory

The central operational boundary in this design log is the boundary between just-in-time sourcing and pre-positioned inventory.

5.1 Just-in-Time Sourcing

Just-in-time sourcing refers to an operational model in which liquidity, settlement assets, collateral, or payout instruments are acquired only when needed. JIT sourcing is likely to work well when direct liquidity pairs are deep, markets remain liquid under stress, settlement assets are widely accepted, finality timing is predictable, local payout is reliable, and internal netting is available.

In such settings, holding inventory may be inefficient. It may impose unnecessary capital costs, custody costs, price risk, and governance burden.

5.2 Pre-Positioned Inventory

Pre-positioned inventory refers to an operational model in which an operator holds an asset before it is needed in order to improve execution reliability, reduce slippage, shorten delay, satisfy customer service levels, or maintain access under stress.

Pre-positioning may become rational when direct pairs are thin, execution size is large, stress-time liquidity disappears, settlement windows do not align, local payout is unreliable, cross-ledger movement is slow, JIT execution failure is costly, customer liquidity buffers must be maintained, or fallback routing is commercially necessary.

6. Boundary Formula

The proposed boundary condition for backend inventory formation is:

JIT-side burden	Inventory-side burden
JIT sourcing cost	Inventory capital cost
Execution failure risk	Custody friction
Market impact	Regulatory / accounting cost
Time-constraint cost	Inventory price volatility risk
Liquidity fragmentation cost	Governance burden
Procurement scarcity risk	
Institutional velocity constraint	
Position-limit cost	
Client liquidity buffer cost	

This formula is a design proposal, not a validated empirical model. It is intended to preserve a research boundary for future testing, not to assert that any asset currently satisfies the condition.

When the JIT-side burden exceeds the inventory-side burden, pre-positioned inventory may become economically rational. This does not prove that any specific asset will be selected. It only defines a boundary at which retained demand may become possible.

7. Inter-Zone Liquidity Boundary Model

The JIT-versus-inventory boundary is especially important when moving from intra-zone workflows to inter-zone workflows.

7.1 Intra-Zone Rails

Some systems are best understood as intra-zone institutional rails. Examples may include Canton-like permissioned networks, tokenized deposit systems, bank-ledger networks, securities settlement systems, repo infrastructure, collateral mobility systems, and controlled institutional synchronization layers.

Within such zones, JIT sourcing, internal netting, tokenized deposit settlement, and permissioned synchronization may reduce the need for volatile bridge assets.

Intra-zone rails can make JIT easier.

7.2 Inter-Zone Boundaries

Inter-zone boundaries occur between currency zones, ledger zones, legal regimes, liquidity pools, stablecoin systems, bank networks, CBDC systems, tokenized deposit networks, and public-chain liquidity environments.

At these boundaries, JIT sourcing may become fragile due to currency mismatch, ledger mismatch, legal and regulatory mismatch, settlement-time mismatch, thin direct pairs, local payout constraints, stress-time liquidity collapse, sanctions and capital controls, issuer or bank-credit differences, and fragmented liquidity venues.

Inter-zone boundaries can break JIT.

8. Macro-Liquidity Patch

This design log also preserves a macro-liquidity hypothesis relevant to future research. This hypothesis helps explain why inter-zone boundaries may remain important even if stablecoins, tokenized deposits, and institutional ledgers become more efficient.

Small or fragile currency islands may be absorbed into stablecoin zones, especially where local currency instability, weak banking access, dollar shortages, or cross-border payment friction make stablecoins attractive. However, major currency, regulatory, banking, CBDC, tokenized-deposit, Big Tech payment, and prime-liquidity zones may persist.

The resulting macro-structure may be summarized as a hypothesis:

Small islands may be absorbed. Major islands may remain. Many major islands may remain. Their liquidity may not be uniform. Uneven liquidity can break inter-zone JIT in certain corridor types, especially where direct-pair depth remains thin under stress. In those corridors, bridges may require not only routing, but liquidity equalization.

This hypothesis must remain open to the opposite outcome: stablecoin network effects, tokenized deposits, CBDC corridors, RTGS modernization, bank-ledger interoperability, or prime-broker internalization may reduce or eliminate the inter-zone frictions that would otherwise make bridge assets relevant.

9. Alternative Infrastructure Maintenance Cost Lens

The inter-zone boundary model must be paired with an alternative-infrastructure maintenance lens. Otherwise, the framework may become biased by treating candidate-asset costs as visible while treating asset-free composite infrastructure costs as invisible.

This section should be read as a comparative cost hypothesis, not as a benefit statement for any candidate asset.

AIMC is not a weapon against alternatives. It is a symmetry device. It asks the same maintenance question on both sides: who keeps the route alive?

Asset-free infrastructure is not cost-free infrastructure. Asset-inclusive infrastructure is not cost-free infrastructure either.

If a candidate asset is bypassed, someone still has to operate the rails, provide liquidity, hold inventory, manage compliance, guarantee local payout, absorb failure risk, and keep the route alive under stress. Examples include stablecoin-native settlement, tokenized deposits plus bank networks, bank APIs plus local payout providers, cross-chain messaging plus stablecoin transfer protocols, prime-broker internalization, exchange internal liquidity, CBDC or RTGS corridors, securities-ledger settlement, and Big Tech payment rails.

9.1 Questions for Asset-Free Composite Infrastructure

1. Who operates the rails?
2. Who provides liquidity?
3. Who holds inventory or buffers?
4. Who absorbs failure costs?
5. Who handles compliance, accounting, AML, and sanctions controls?
6. Who guarantees local payout?
7. Who maintains stress-time access?
8. Who earns enough to keep the route alive?

9.2 Questions for Asset-Inclusive Bridge Configuration

1. Who holds the bridge asset?
2. Who hedges volatility?
3. Who absorbs mark-to-market risk?
4. Who handles custody, accounting, audit, and regulatory treatment?
5. Who maintains sufficient depth under stress?
6. Who bears the cost if banks, customers, or counterparties reject exposure to the asset?
7. Who proves that the asset's apparent depth is real at large ticket sizes?
8. Who earns enough to justify carrying the asset rather than sourcing or bypassing it?

10. Comparative Cost Model

The comparative model should evaluate the cost of maintaining a bridge against the cost of maintaining complexity.

Comparison layer	Asset-inclusive bridge configuration	Asset-free composite infrastructure
Core burden	Holding and maintaining bridge liquidity	Maintaining multiple rails and coordination layers
Visible cost	Inventory, custody, hedging, regulatory/accounting treatment	Bank APIs, stablecoin liquidity, payout providers, messaging layers, compliance
Hidden risk	Volatility, hedging expense, mark-to-market burden, custody constraints, accounting/audit friction, bank/customer rejection, asset-specific rejection, depth illusion	Fragmented liquidity, issuer/redemption risk, local payout failure, multi-party coordination
Stress condition	Can the asset remain deep and usable under stress?	Can the composite route remain available and liquid under stress?
Audit test	Does holding the asset reduce total cost, risk, or complexity?	Does bypassing the asset avoid cost without recreating equal or greater maintenance burden?

10.1 Asset-Inclusive Configuration

A candidate asset-inclusive configuration may be represented as inventory capital cost plus volatility and hedging cost, custody and technical friction, regulatory and accounting cost, LP/MM inventory maintenance cost, and depth-maintenance cost, minus slippage reduction, execution-failure reduction, reachability improvement, and composite-infrastructure simplification.

10.2 Asset-Free Composite Infrastructure

An asset-free composite configuration may be represented as multi-rail maintenance cost plus stablecoin or tokenized-deposit liquidity cost, bank API or payout provider cost, cross-chain messaging or bridge cost, custody/redemption/issuer risk, jurisdiction-specific compliance cost, local payout cost, stress-time alternative-route cost, liquidity fragmentation cost, failure/SLA cost, and multi-party coordination cost.

A candidate asset may have operator-layer relevance only if the asset-inclusive configuration has lower total cost, risk, or complexity than the asset-free composite alternative. Even then, retained demand requires evidence that operators actually hold the asset as inventory, collateral, margin, liquidity buffer, or fallback liquidity.

11. XRP as Stress-Test Case

XRP may be used as a stress-test case for this framework because it is often discussed as a bridge asset, liquidity asset, cross-border settlement asset, or institutional infrastructure component. However, XRP is not treated as the expected winner, default bridge, inevitable liquidity layer, or required settlement asset.

Under this framework, XRP retained demand would require evidence that infrastructure operators hold XRP because doing so improves inter-zone execution, reduces slippage, lowers failure risk, improves reachability, simplifies composite infrastructure, or strengthens fallback liquidity.

Relevant evidence may include XRP inventory held by liquidity providers or market makers, XRP collateral or margin use by prime brokers or institutional counterparties, XRP liquidity buffers held by payment operators or exchanges, deep and durable XRP/stablecoin or XRP/fiat liquidity, large-order execution support involving XRP, XRP use in rebalancing between zones, and XRP removal sensitivity in cost, delay, reachability, or failure rate.

11.1 Conditions Under Which the XRP Hypothesis Should Be Downgraded

The XRP hypothesis should be downgraded if XRP is technically and institutionally available but still fails to appear as operator inventory, collateral, margin, liquidity buffer, or fallback liquidity across relevant domains.

It should also be downgraded if XRP removal produces no measurable deterioration in cost, delay, slippage, reachability, failure rate, collateral efficiency, or infrastructure complexity.

A further downgrade is warranted if RLUSD, USDC, tokenized deposits, bank APIs, prime-broker internalization, or other rails solve the same workflow without XRP and without recreating equal or greater maintenance burden.

The strongest downgrade case would be persistent non-selection across payments, prime brokerage, custody, treasury, market-making, collateral, and rebalancing domains after relevant infrastructure and regulatory preconditions are sufficiently mature.

12. Evidence Requirements

Backend retained demand requires more than adoption narratives. Observed balances alone are not sufficient. Inventory evidence requires purpose documentation or independently auditable context showing that the asset is held for operational inventory, collateral, margin, liquidity buffering, rebalancing, or fallback access rather than passive custody, speculation, legacy holding, or customer deposits.

Evidence category	Examples
Inventory evidence	Operator-held inventory, LP/MM pre-positioning, exchange settlement buffers, prime-broker inventory, payment-corridor liquidity, treasury rebalancing reserves, fallback liquidity lines.
Collateral and margin evidence	Use as institutional collateral or margin, acceptance by counterparties, deterioration when removed from collateral sets, integration into prime-broker or custody workflows.
Liquidity-depth evidence	Deep order books, low large-order slippage, stress-time depth resilience, durable stablecoin pairs, distributed LP participation, working-float availability.
Removal sensitivity	Higher cost, delay, slippage, failure rate, reduced reachability, weaker payout reliability, worse rebalancing, lower collateral efficiency, increased infrastructure complexity when the asset is removed.

Without removal sensitivity, asset necessity remains weak.

13. Non-Selection Evidence

Non-selection evidence is essential because a retained-demand audit must be able to lower a hypothesis as well as raise it. Non-selection evidence is produced when a candidate asset is not selected despite apparently favorable conditions.

This section requires a careful distinction between non-selection, technical unavailability, and negative selection. Non-selection should not be inferred merely because an asset is absent from a workflow. Absence may reflect technical unavailability, immaturity, missing integrations, legal delay, or lack of access. Non-selection becomes more meaningful only when the asset is technically available, operationally accessible, institutionally feasible, and still not used or held.

Negative selection is a stronger form of non-selection. It occurs when an operator appears to choose a more expensive, slower, or more complex route because non-economic constraints outweigh the candidate asset's apparent cost advantage. These constraints may include regulation, accounting treatment, bank counterparty limits, custody policy, customer rejection, sanctions exposure, internal risk limits, or governance rules.

The threshold for treating non-selection as hypothesis-lowering rather than immaturity has not yet been operationalized. Future work should specify observation windows, domain coverage, institutional preconditions, and downgrade levels.

14. Evidence-Neutral Audit Posture

The audit posture for this design log is evidence-neutral. This section is not a rhetorical disclaimer; it is a methodological requirement.

Universal OS is not a device for making XRP win. It is a device for identifying where, why, and on whose balance sheet XRP demand would appear if XRP wins. It is also a device for recognizing where and why XRP was not selected if it does not win.

This principle should generalize beyond XRP. The same discipline should apply to any institutionally connected digital asset. The framework should not be soft on XRP. It should not be soft on alternatives. It should apply equal rigor to confirmation and falsification. Attractive structures should be treated as maps, not conclusions.

15. Bias Controls

Caution terms	Safer terms
inevitable; already decided; only this asset; alternatives are impossible; only a matter of time; structurally it must happen	candidate; hypothesis; conditional; unconfirmed; audit target; evidence pending; alternative comparison required; non-selection would lower the thesis

The purpose of these controls is to prevent the framework from becoming an expectation-preserving device.

16. Standard Audit Protocol

1. Identify the relevant zone or island.
2. Determine whether the workflow is intra-zone or inter-zone.
3. Assess whether JIT sourcing is likely to work or break.
4. Identify where liquidity is thick and where it is thin.
5. Determine who maintains the asset-free bridge.
6. Assess whether the asset-inclusive route compresses total cost, risk, or complexity.
7. Look for actual inventory, collateral, margin, liquidity buffers, or fallback liquidity.

8. Test whether removing the asset degrades outcomes.
9. If the asset is not selected despite favorable conditions, assess non-selection evidence.
10. Define what evidence would raise, lower, or veto the hypothesis.

17. Relationship to the Retained-Demand Audit Series

Paper	Connection
Paper 1	Focused on retained demand by type. This design log focuses on the boundary conditions under which retained demand may arise at the operator layer.
Paper 2	Distinguished expansion from closure. This design log preserves the distinction by emphasizing that infrastructure expansion does not automatically close into asset-level retained demand.
Paper 3	Separated sizing from price thesis. This design log reinforces that valuation inputs remain weak until inventory, collateral, liquidity buffers, or removal sensitivity are evidenced.
Paper 4	Examined whether infrastructure expansion leads to bypass, compression, or amplification. This design log extends that inquiry to operator-layer cost compression and asset-free composite infrastructure.
Paper 5	Emphasized evidence before sizing. This design log applies that discipline to backend retained demand after user abstraction.
Paper 6	Examined where retained demand may appear after user abstraction. This design log asks when it appears, why it appears, and what should be concluded if it does not appear.

18. Possible Future Paper Structure

If this design log later develops into a full paper, a possible structure may be the following. This is included only to preserve design continuity and should not be read as a launch decision:

1. Introduction
2. From User Abstraction to Backend Retained Demand
3. JIT Sourcing Versus Pre-Positioned Inventory
4. Intra-Zone Rails and Inter-Zone Liquidity Boundaries
5. Alternative Infrastructure Is Not Free Infrastructure
6. Operator-Layer Cost Compression Model
7. Evidence Requirements: Inventory, Collateral, Liquidity Buffers, Removal Sensitivity
8. Non-Selection Evidence
9. XRP as Stress-Test Case
10. Implications for Evidence-Gated Sizing
11. Conclusion

19. Current Non-Launch Rationale

Paper 7 has not been launched for the following reasons: the current document is a design log rather than a full paper; the structure is promising but evidence remains incomplete; more observation is needed regarding operator inventory, collateral use, liquidity depth, and removal sensitivity; persistent non-selection evidence has not yet been fully evaluated; and the framework should remain in a design-log state until the empirical and comparative boundary conditions are clearer.

Design Log / Research Note / Timestamped Archive

20. Summary

This design log should be read as a research-design artifact. It develops a possible boundary model for backend retained demand after user abstraction. It asks when infrastructure operators must move from just-in-time

sourcing to pre-positioned inventory, collateral, liquidity buffers, or fallback liquidity lines. The model distinguishes intra-zone institutional rails, where JIT and netting may work efficiently, from inter-zone boundaries, where currency, ledger, jurisdiction, settlement-time, local-payout, and liquidity-depth frictions may make JIT fragile.

It further argues that asset-free composite infrastructure is not free infrastructure. If a candidate asset is bypassed, someone still has to maintain the rails, provide liquidity, hold inventory or buffers, manage compliance, guarantee payout, absorb failure risk, and keep the route alive under stress. The relevant comparison is therefore not whether a candidate asset can be bypassed, but whether the asset-inclusive configuration has lower total cost, risk, or complexity than the asset-free composite alternative.

XRP is used as a stress-test case, not as a predetermined winner. XRP retained demand would require evidence that operators hold XRP because it reduces cost, delay, slippage, failure rate, reachability risk, or infrastructure complexity. If operators do not hold XRP despite favorable boundary conditions, persistent non-selection should be treated as evidence against the hypothesis.

21. Short Form

Small islands may be absorbed. Major islands may remain. Many major islands may remain. Their liquidity may not be uniform. Where direct liquidity remains thin, bridges may need more than routing. They may need liquidity equalization.

A candidate asset becomes relevant only if holding it allows infrastructure operators to reduce total cost, risk, complexity, slippage, delay, or failure relative to JIT sourcing or asset-free composite infrastructure. Whether any specific asset satisfies these conditions remains unconfirmed.

Use without inventory is weak retained demand. Inventory appears only when holding beats sourcing. Non-selection matters when favorable conditions exist but operators still refuse to hold the asset.

22. Suggested Archive Metadata

Field	Suggested value
Suggested OSF component	Phase II Materials
Suggested OSF folder	design-logs / paper-7-candidate-a-design-log-v0.1
Suggested document type	Design Log / Research Note / Timestamped Archive
Suggested public label	Not a full paper draft. Paper 7 has not been launched.
Suggested use	Research continuity, priority timestamping, future design work, external review baseline
Suggested non-use	Investment thesis, XRP price thesis, proof of asset demand, or Paper 7 draft

23. Final Status

Item	Status
Document	Paper 7 Candidate A Design Log v0.1
Status	Design Log v0.1 — Revised Candidate / Research Note / Timestamped Archive
Series	Retained-Demand Audit Series — Phase II Materials
Paper 7	Not launched
Use	Research continuity, timestamping, OSF archive, future design work, external review baseline
Do Not Use As	XRP price thesis, Paper 7 draft, investment claim, proof of asset demand, or evidence that any specific asset has already been selected
Focus	Boundary modeling of backend liquidity mechanics

Keywords

User abstraction; backend retained demand; just-in-time sourcing; pre-positioned inventory; operator-layer cost compression; alternative infrastructure maintenance cost; inter-zone liquidity boundaries; non-selection evidence; removal sensitivity; inventory formation; liquidity buffers; collateral; margin; digital assets; stablecoins; tokenized deposits; XRP; institutional settlement; financial market infrastructure.

Author Note

This document is presented as a design log and research note within the Retained-Demand Audit Series. It is intended to preserve the research sequence after Paper 6 and to provide a timestamped basis for possible future work on backend retained demand, operator-layer cost compression, inventory formation, and non-selection evidence.

It does not provide investment advice, a price target, or a recommendation to buy, sell, or hold any digital asset. XRP appears only as a stress-test case in a broader audit framework.

Short Description

A Phase II design log extending retained-demand analysis after user abstraction. The note develops a boundary model for when backend operators may need to move from just-in-time sourcing to pre-positioned inventory, collateral, liquidity buffers, or fallback liquidity lines, and it defines how non-selection evidence should lower retained-demand hypotheses when favorable conditions exist but operators still refuse to hold the asset.